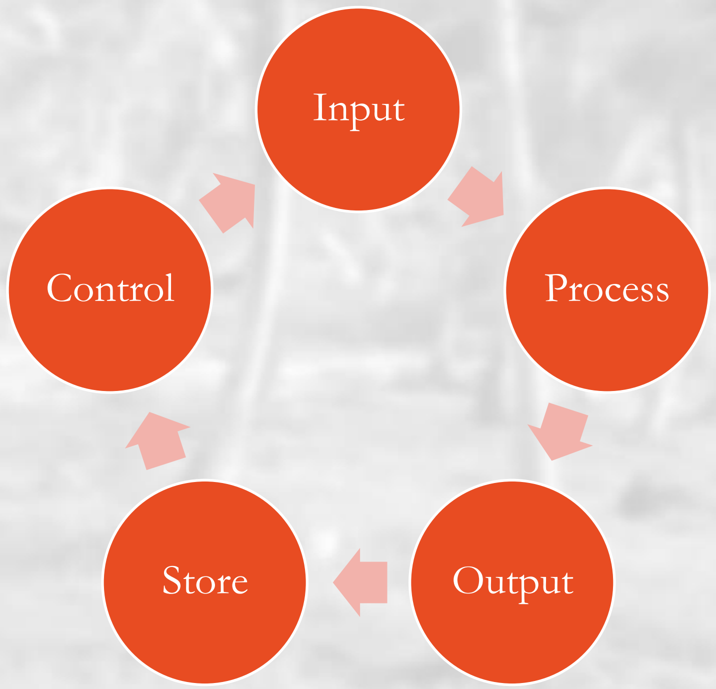
Operating System

By Mohamed Gamal

Basic Computer Configuration

– The computer is an electronic machine that performs the following five basic operations:

- Input
- Process
- Output
- Store
- Control



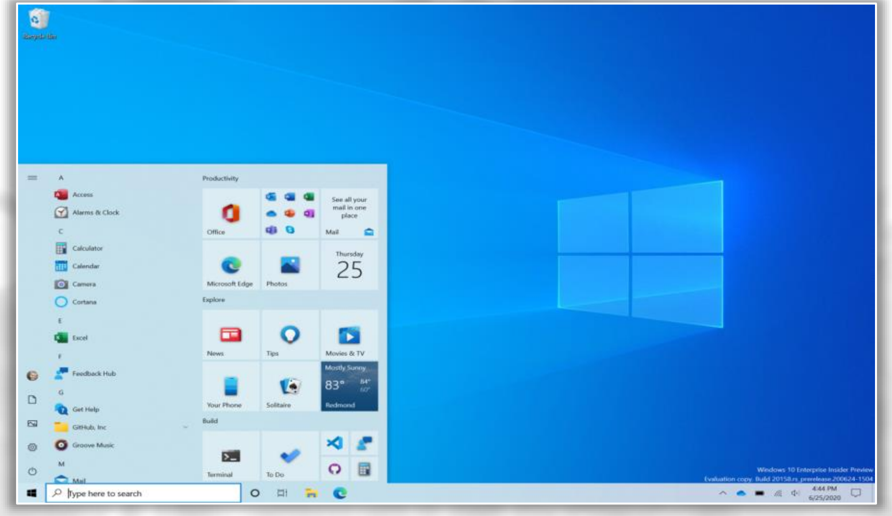
What's An Operating System (OS)?

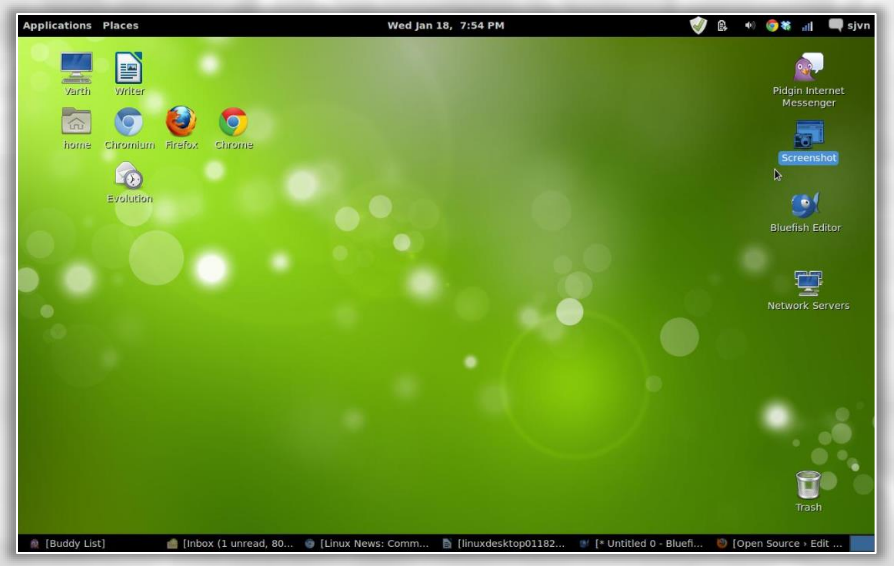
– A program that acts as an intermediary between a user of a computer and the computer hardware.

– Operating System goals:
  - Execute user programs
  - Use the computer hardware in an efficient manner
  - Make user's life easier ;)

– Operating System examples:
  - Windows
  - Unix / Linux
  - MacOS

OSes User Interfaces

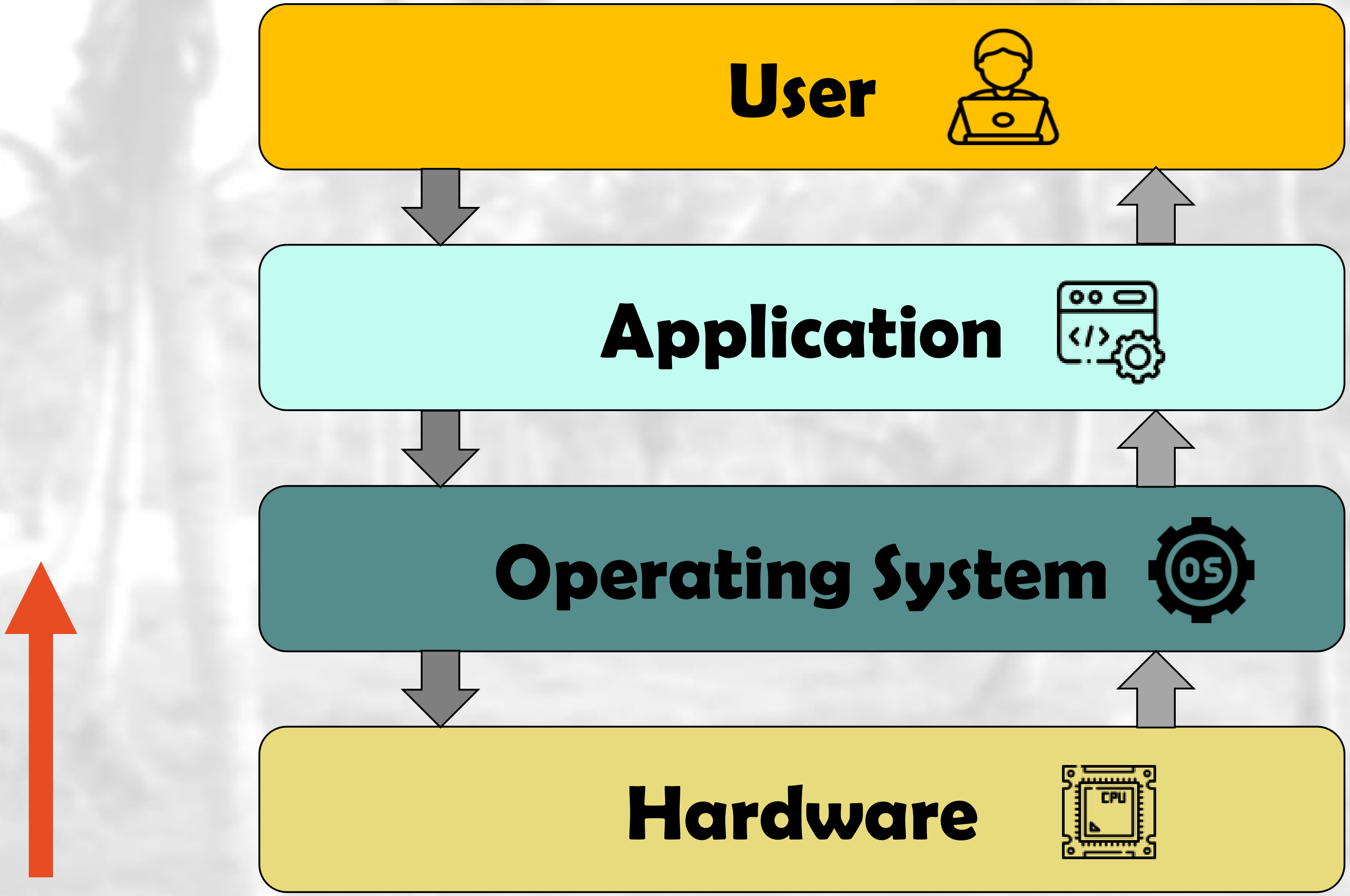




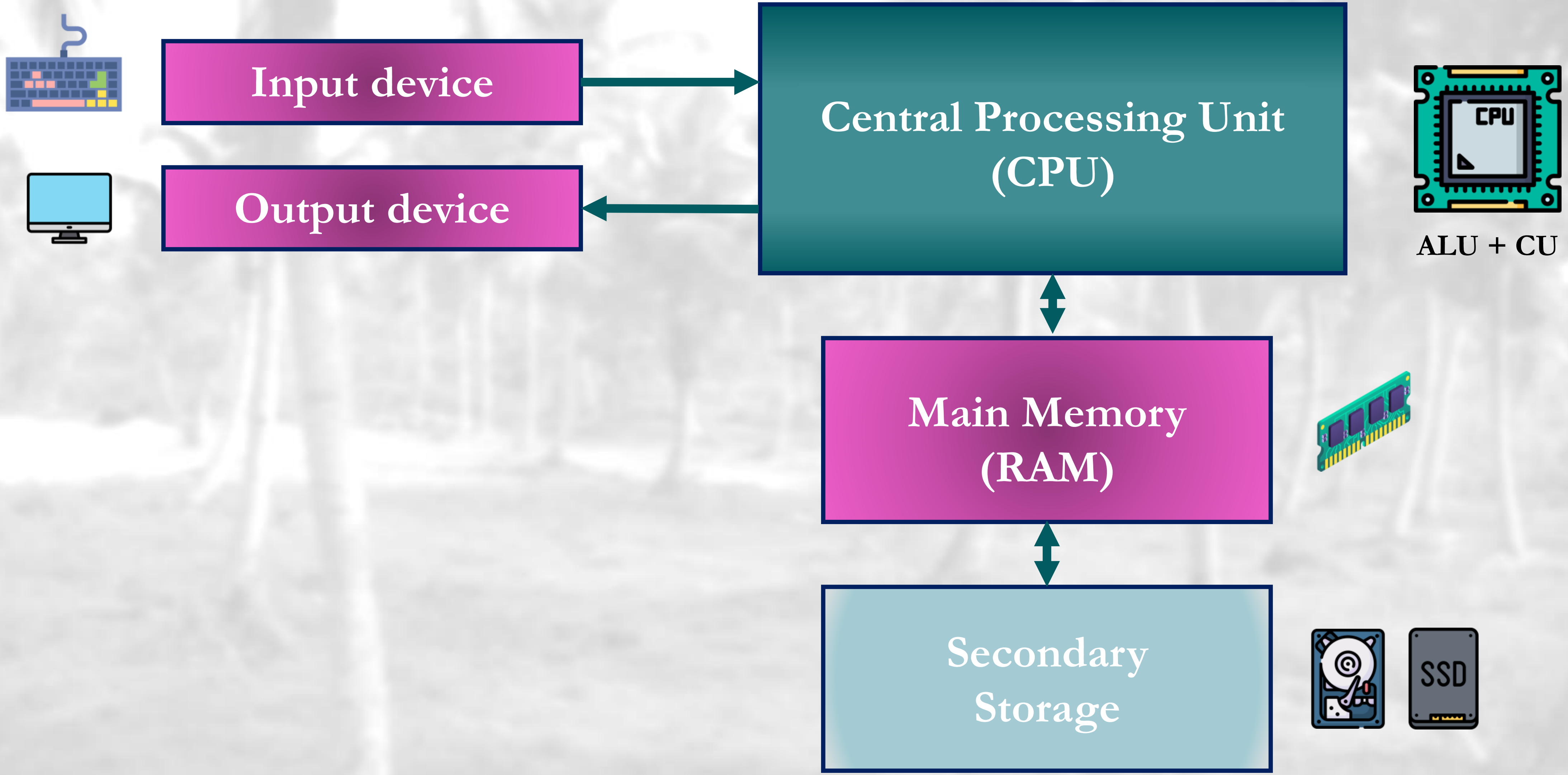
History of Operating Systems

- Operating systems were first *developed in the late 1950s* to manage tape storage.
- The General Motors Research Lab implemented the first OS in the early 1950s for their *IBM 701*
- In the late 1960s, the first version of the *Unix OS* was developed
- The first OS built by Microsoft was *DOS*, *built in 1981*.
- The present-day popular OS Windows first came to existence in 1985 when a GUI was created and paired with MS-DOS.

Computer System Components

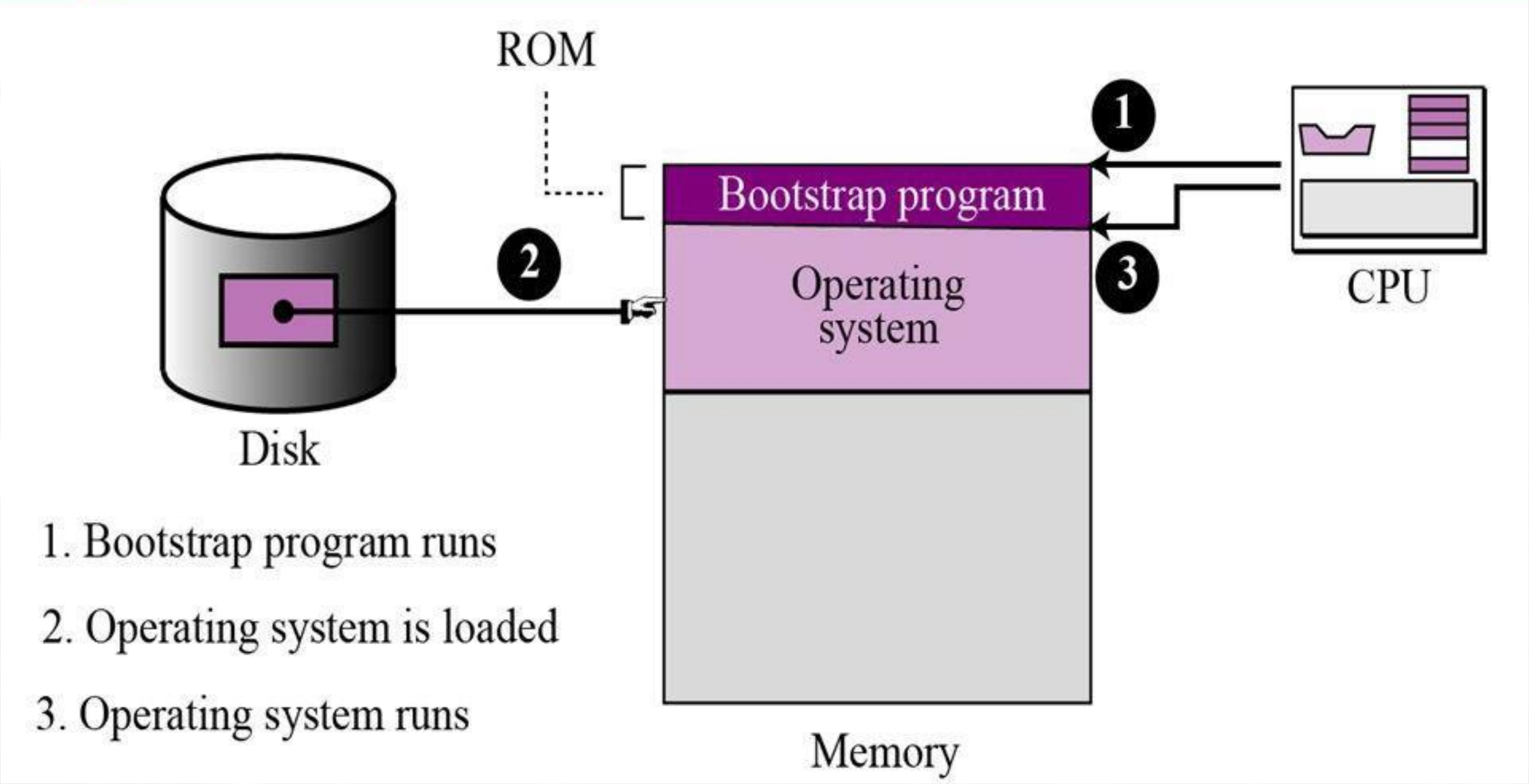


1) Computer Hardware



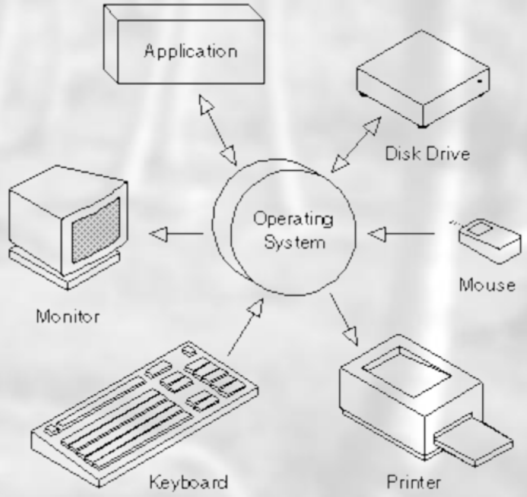

AMD
RYZEN
AMD Ryzen 7 7800X3D
SSD PLUS
Solid State Drive
SanDisk
V-NAND SSD
SSD 980
1TB
SAMSUNG
EXTREME
HDMI
WI-FI

Bootstrap Process



2) Operating System

- OS controls and coordinates the use of hardware among various application programs for users
  - It manages and allocates the resources (e.g., CPU, Storage, …, etc.)
  - It controls execution of user programs and operations of I/O devices
- OS is a **Resource allocator** and **Control program**
- **Kernel** — the one program running at all times



3) Application Programs

– Video Games
– Web browsers
– Spread sheets
– Word processors
– Compilers and Assemblers
– …

3) Application Programs

A Source Code/Text Editor is a type of software used for writing and editing plain text, particularly source code for computer programs. These editors provide a variety of features that help developers write, modify, and navigate through code efficiently.

- Notepad (System Application)
- Notepad++
- Visual Studio Code
- Sublime Text
- Atom
- Etc.

3) Application vs. System Programs

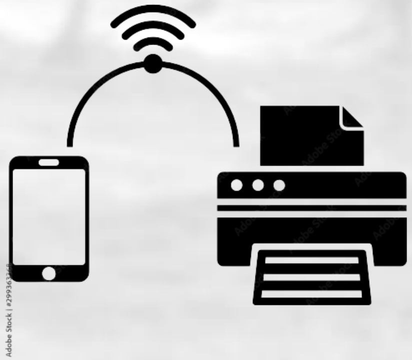
– System programs, also known as operating system utilities or system utilities, are software programs that are shipped with the OS and are an integral part of the operating system.

- Status Information
- File Modification
- Drivers
- …

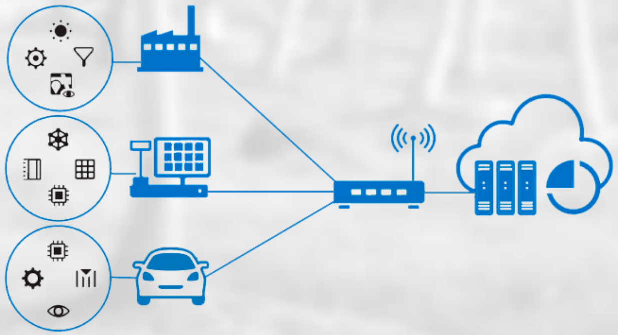
4) Users

- People (e.g., normal users)
- Machines (e.g., servers, embedded systems … etc.)
- Other Computers (e.g., remote or networked devices … etc.)

Users



Machines

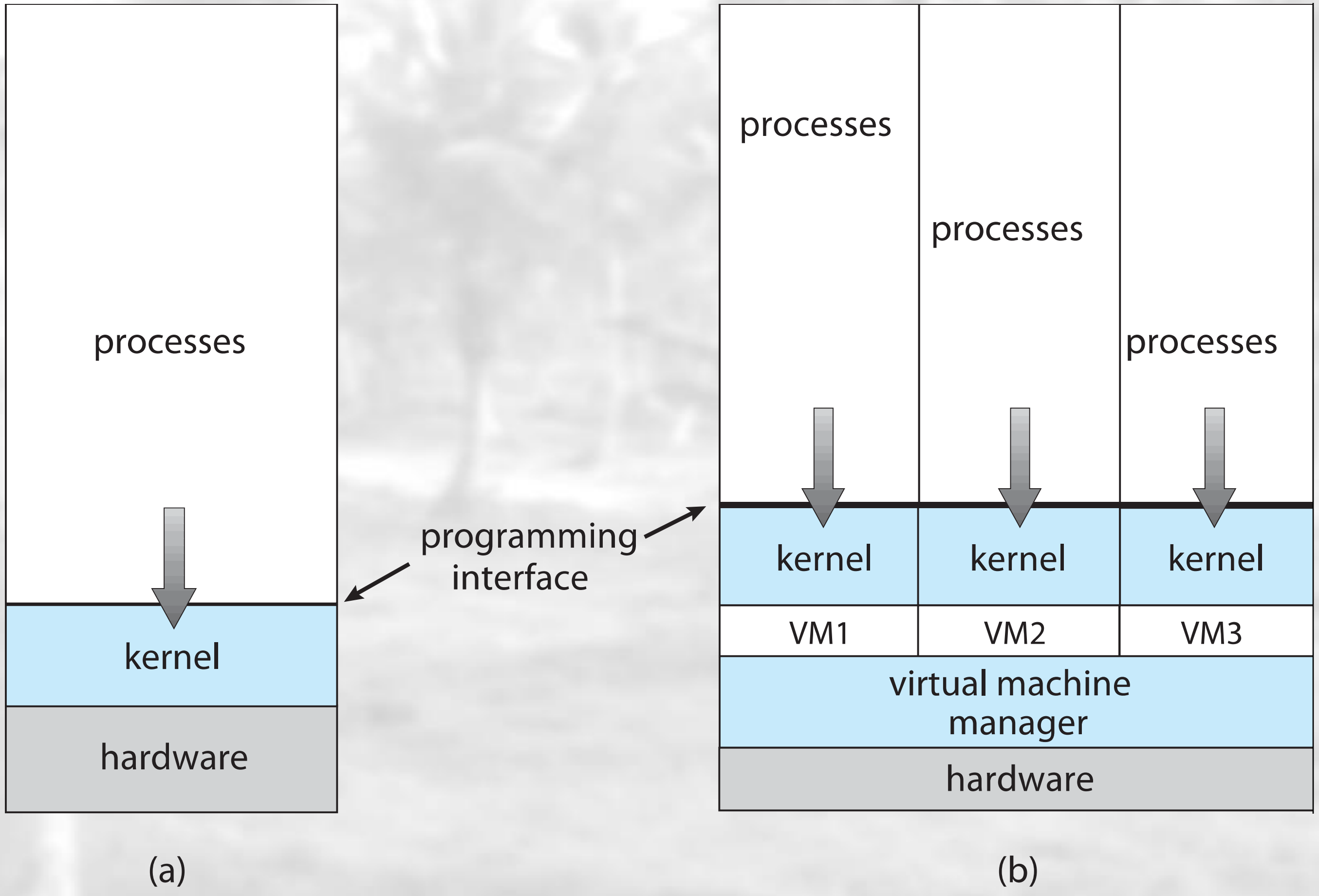


Other Computers

Virtualization

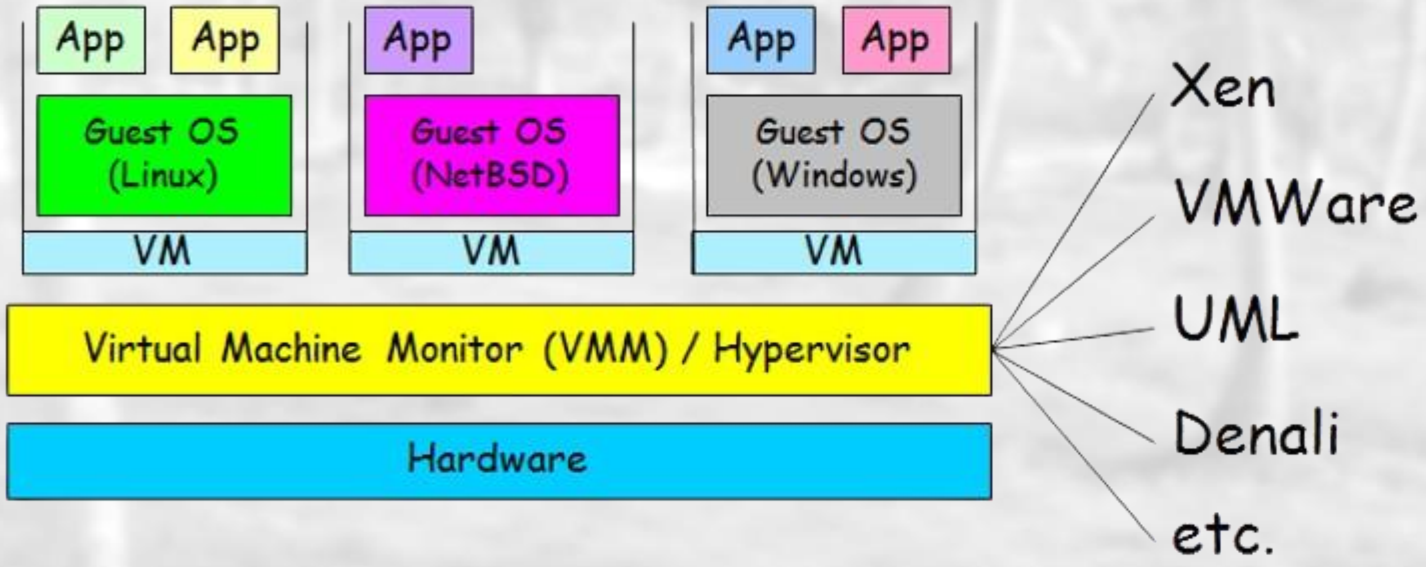
Virtualization

- Allows operating systems to run applications within other OSes.



Virtualization (Cont.)

- VM technology allows multiple virtual machines to run on a single physical machine.
- Virtualization is delivered via so-called virtual machines (VMs).
- The host OS provides a **hypervisor**, which manages the access to the hardware.

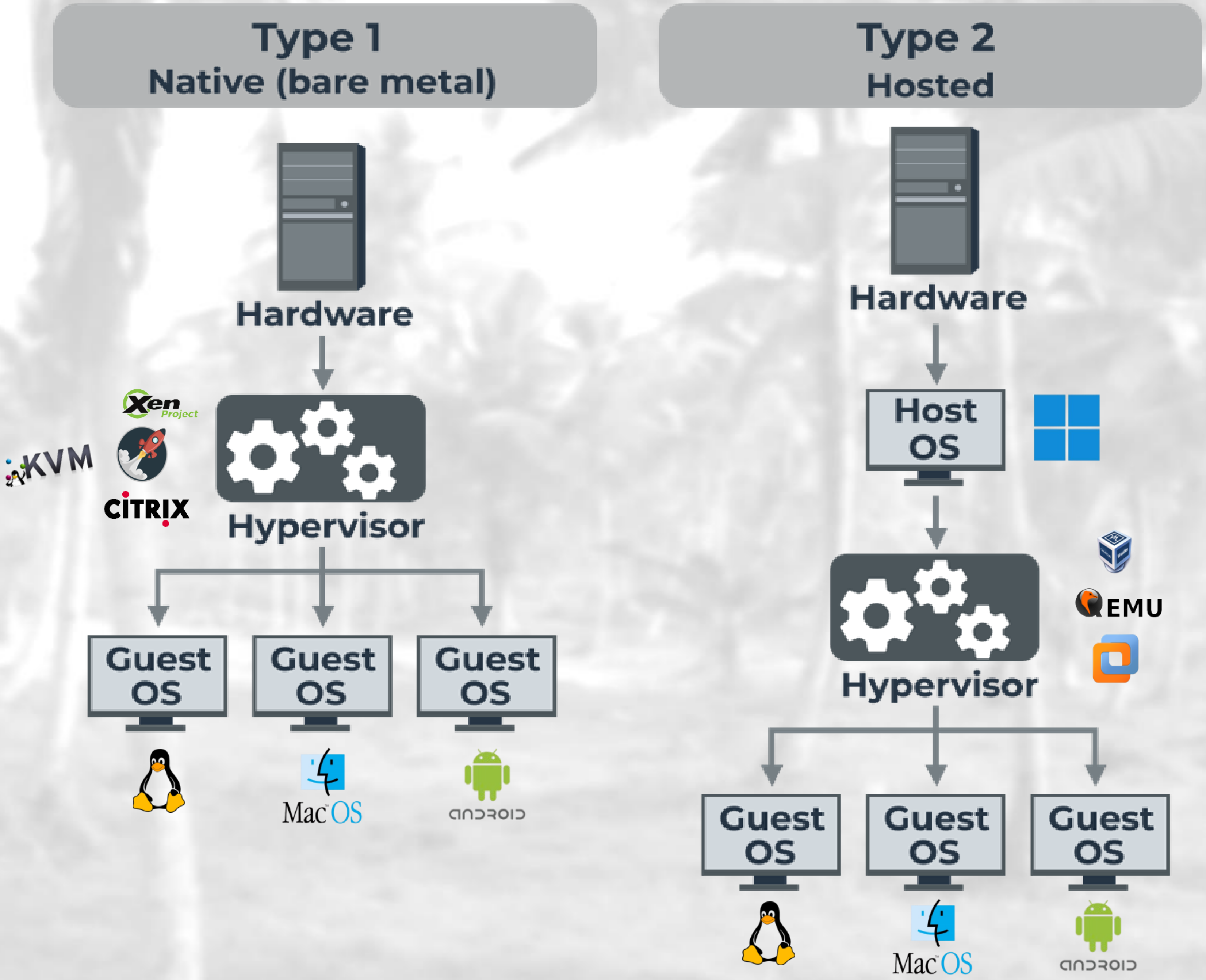


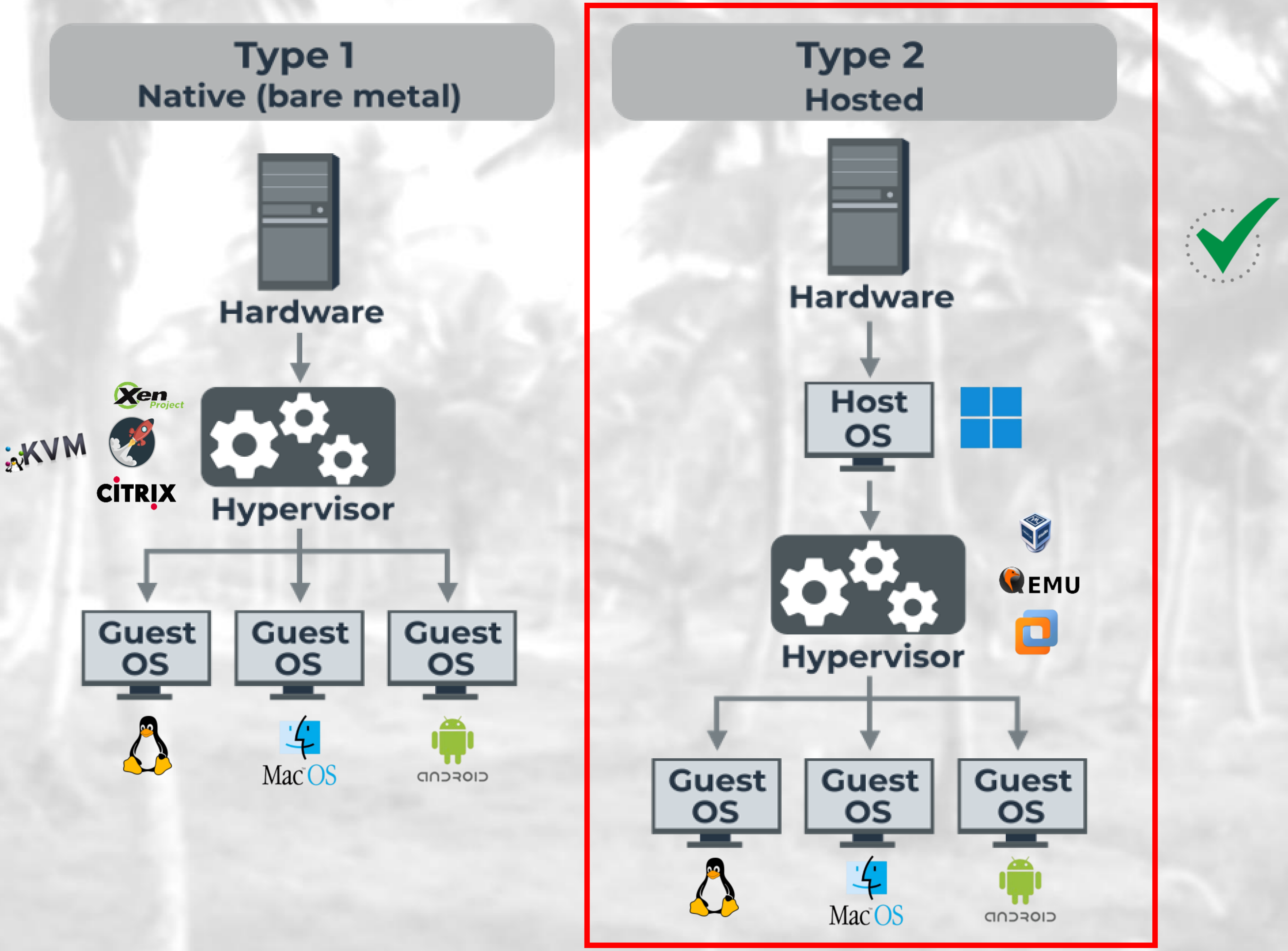
What is Hypervisor?

- The hypervisor is a software component that manages multiple virtual machines in a computer.
- It ensures that each virtual machine gets the allocated resources and does not interfere with the operation of other virtual machines. There are two types of hypervisors.

Type 1 hypervisor (bare-metal)

Type 2 hypervisor (hosted)


Type 1
Native (bare metal)
Hardware
Xen Project
KVM
CITRIX
Hypervisor
Guest OS
Guest OS
Guest OS
Mac OS
android
Type 2
Hosted
Hardware
Host OS
Hypervisor
EMU
Guest OS
Guest OS
Guest OS
Mac OS
android


Type 1
Native (bare metal)
Hardware
Hypervisor
KVM
CITRIX
Guest OS
Guest OS
Guest OS
Mac OS
android
Type 2
Hosted
Hardware
Host OS
Hypervisor
EMU
Guest OS
Guest OS
Guest OS
Mac OS
android

Virtualization Softwares

vmware

[Link]

ORACLE VM VirtualBox

[Link]

Different Types of Virtualization

1) Server Virtualization

- It is a process that partitions a physical server into multiple virtual servers.
- It is an efficient and cost-effective way to use server resources and deploy IT services in an organization.
- Without server virtualization, physical servers use only a small amount of their processing capacities.

2) Application Virtualization

- It pulls out the functions of applications to run on operating systems other than the operating systems for which they were designed.
- For example, users can run a Microsoft Windows application on a Linux machine without changing the machine configuration.

What are VMs Used for?

- Cloud computing
- Software testing
- Malware Investigations
- Disaster Management

https://ae.godaddy.com/hosting/vps-hosting

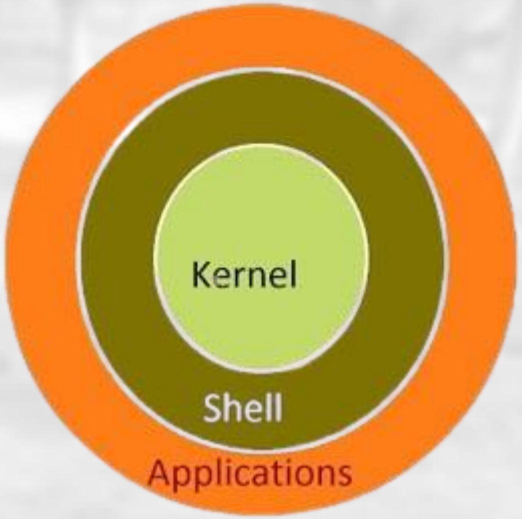
Command Line

Different types of command line interfaces

- Linux-based OS
- Microsoft Windows
- macOS (Apple computers)
- iOS (Apple smartphones/tablets)
- Android
- etc.

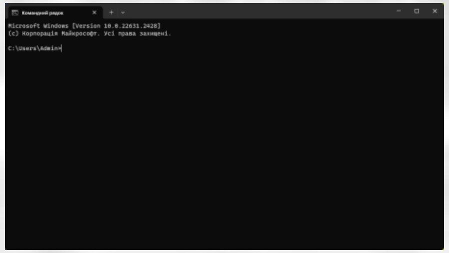
Different types of command line interfaces

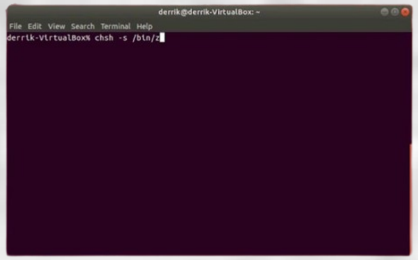
- **Shell:** a command-line interface for interacting with the OS.
- **Terminal:** the application to access the Shell.
- **CMD:** the traditional/original Windows command-line tool.
- **PowerShell:** a more advanced Windows shell with scripting capabilities.

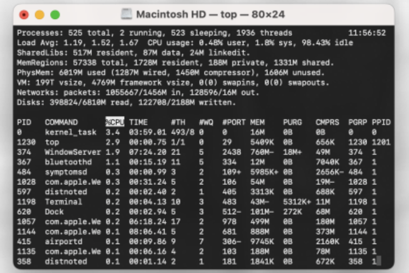


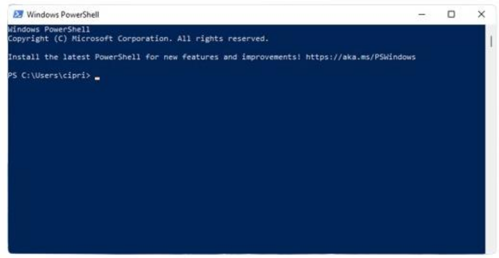
Command Line Interfaces (CLI)

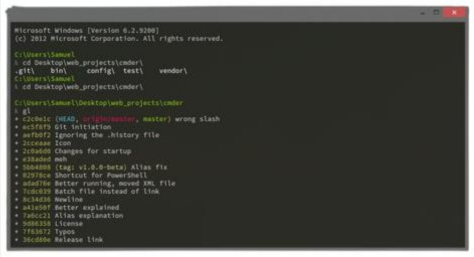
Powershell | **Command Prompt** | **CMDER** | **Bash** | **macOS Terminal**





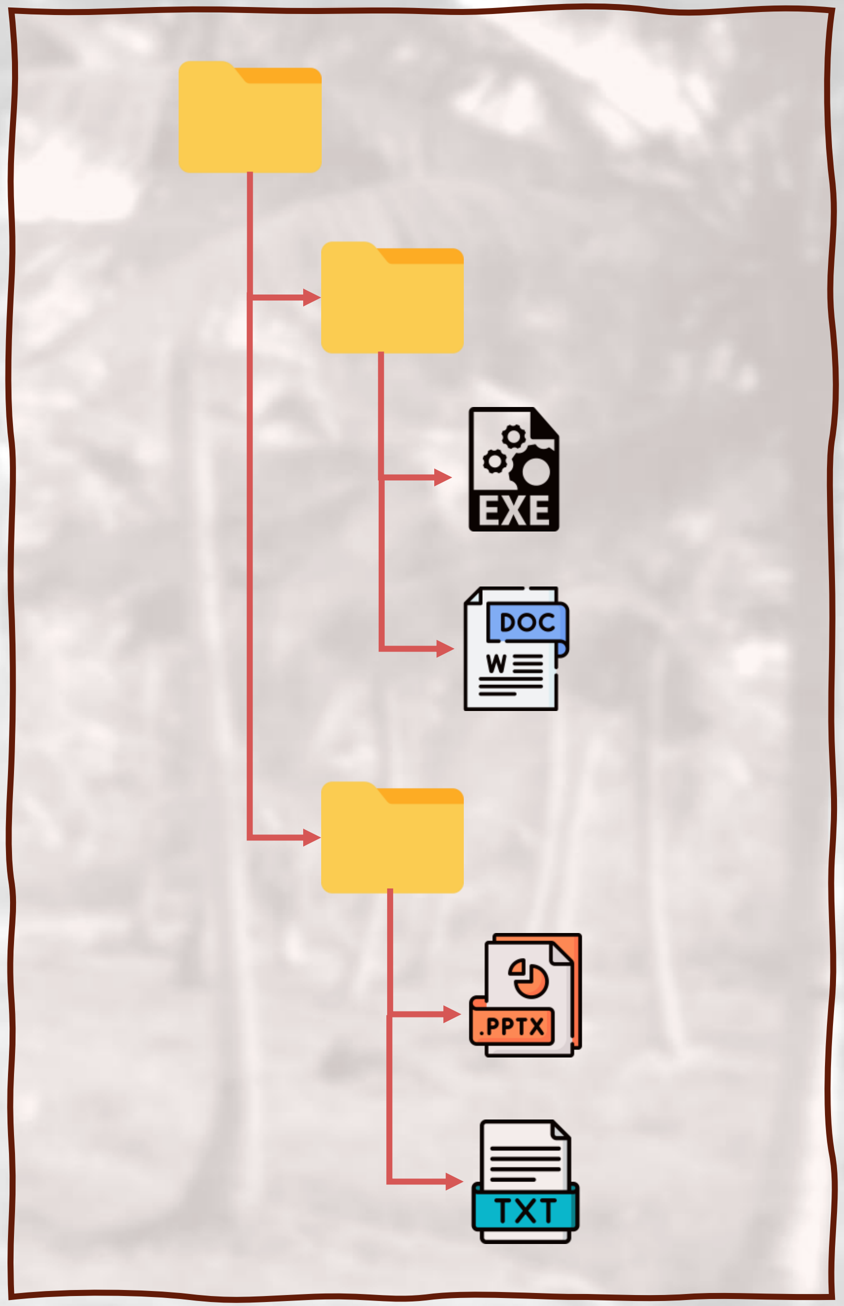






Directories and Files

- Most OSs organize their files in a hierarchical form with files organized inside directories (aka folders).
- There're many application-specific files, with their own file formats.
- It's up to the programmer to define and identify if they require a custom file format for their application or if they can leverage a standard or common file format (e.g., c, py, …, etc.).



Directory Namespace

Key Reasons to Use the Command Line

- **Efficiency:** Execute tasks quickly and efficiently.
- **Automation:** Script complex or repetitive tasks.
- **Powerful Tools:** Access advanced utilities and features.
- **Resource Light:** Minimal CPU and RAM usage.
- **Remote Management:** Administer systems remotely via SSH.
- **Customization:** Personalize with scripts and commands.
- **Precision:** Gain detailed control and options.
- **Troubleshooting:** Diagnose issues with logs and error messages.
- **Learning & Skills:** Enhance technical proficiency and system understanding.

Some Use Cases

- Move all text files in sub-directories
- Delete files with specific extensions
- Change settings more easily
- Create many directories at once
- etc.

Thank You!

End of today's session